



King Abdul Aziz University

Faculty of Computing & Information Technology Computer Science Department

Final Exam, Monday, August 13th, 2018
CPCS-223, Analysis and Design of Algorithm

Time: 120 minutes

Student ID: _____

Student Name: _____

***Section: _____

Instructor:

☐ Dr Umair

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Question 01

What will be the value of

$$\sum_{j=2}^n j$$

- a) $(n(n+1)/2) - 1$ b) $\frac{n^2}{2} + \frac{n}{2} - 1$
c) $(n^2 + n - 2) / 2$ d) All of above

Question 02

According to the definition, which condition(s) is / are not correct for Omega-Notation?

- a) $f(n) > 0$ b) It is tight bound
c) c must be a positive constant d) It describes Upper Bound only

Question 03

If $\frac{1}{2}n^2 - \frac{1}{2}n = \Theta(n^2)$ for $c_1 = \frac{1}{4}$ and $c_2 = \frac{1}{2}$, what will be the value of n_0 ?

- a) $n_0 > 2$ b) $n_0 \geq 2$
c) $n_0 < 2$ d) $n_0 \leq 2$

Question 04

“Loop Invariant” is used for defining

- a) Algorithm’s Efficiency b) Algorithm’s Correctness
c) Algorithm’s Design d) Both a & b

Question 05

“Loop Invariant” can be defined for

- a) Iterative Algorithms b) Recursive Algorithm
c) Both a & b d) None of above

Question 06

“Loop Invariant” can be written in

- a) Descriptive Statement b) Mathematical Expression
c) Both a & b d) None of above

Question 07

“At the start of each iteration of the for loop the subarray $A[1 \dots j-1]$ consist of the elements originally in $A[1 \dots j-1]$ but in sorted order.” is the loop invariant for:

- a) Merge Sort b) Insertion Sort
c) Quick Sort d) None of above

Question 08

“Recurrence” is described as

- a) Equation / Equality b) Inequality
c) Both a & b d) None of above

Question 09

“Recurrence” is used for describing the running time of:

- a) Iterative Algorithms b) Recursive Algorithm
c) Both a & b d) None of above

Question 10

Which of the following is / are correct for “Recurrence” $T(n) = a T(n/b) + f(n)$

- a) $T(n/b)$ define the running time of a sub-problem
- b) “a” are the number of sub-problems
- c) $f(n)$ describe the running time of all statements in algorithm except recursive calls.
- d) All of above

Question 11

Which of the following is / are incorrect for “Substitution Method”

- a) Can solve upper and lower bounds simultaneously
- b) Powerful method to solve recurrences
- c) demands practice / expertise
- d) Use Mathematical Induction to prove the guess

Question 12

Which condition is incorrect for general recurrence equation?

- a) $a \geq 1$
- b) $b > 1$
- c) $b < 1$
- d) None

Question 13

What is the value of “b” for the recurrence equation $T(n) = T(2n/3) + O(n)$

- a) $2/3$
- b) $3/2$
- c) 3
- d) $1/3$

Question 14

What is the value of “q” for $q = \lfloor (p + r)/2 \rfloor$, if $p = 5$ and $r = 14$

- a) 9
- b) 10
- c) 9.5
- d) None

Question 15

Which of the following is not a method to solve the recurrence?

- a) Mathematical Induction
- b) Substitution
- c) Master Theorem
- d) None

Question 16

Which of the following methods use “guess” to solve the recurrence?

- a) Recursion Tree
- b) Substitution
- c) Master Theorem
- d) None

Question 17

How many steps are there in “Mathematical Induction” to prove the guess?

- a) 2
- b) 3
- c) 4
- d) None

Question 18

What will be the solution of the following recurrence?

$$T(n) = T(n - 1) + n$$

- | | |
|----------|--------------|
| a) n | b) $\lg n$ |
| c) n^2 | d) $n \lg n$ |

Question 19

Which part of the “General Recurrence” $T(n) = a T(n/b) + f(n)$ will become the root of the tree?

- | | |
|-------------|------------|
| a) $T(n/b)$ | b) n / b |
| c) $T(n)$ | d) $f(n)$ |

Question 20

What will be the solution of the following recurrence? $T(n) = 2T(\lfloor n/2 \rfloor + 17) + n$

- | | |
|----------|--------------|
| a) n | b) $\lg n$ |
| c) n^2 | d) $n \lg n$ |

Question 21

The problem size of a sub-problem in the “General Recurrence” $T(n) = a T(n/b) + f(n)$ is

- | | |
|------------|--------|
| a) n | b) b |
| c) n / b | d) a |

Question 22

The cost of the “recursion tree” can be determine by the sum of the cost of:

- | | |
|-------------------------|-----------------|
| a) all nodes | b) all levels |
| c) all sub-trees & root | d) all of above |

Question 23

In “recursion tree” method, verifying the result using “Substitution Method” is optional.

- | | |
|---------|----------|
| a) TRUE | b) FALSE |
|---------|----------|

Question 24

How we can determine the “height” of a binary tree?

- | | |
|----------------|------------|
| a) n | b) $\lg n$ |
| c) $\lg n + 1$ | d) None |

Question 25

How we can determine the “number of levels” of a binary tree?

- | | |
|----------------|------------|
| a) n | b) $\lg n$ |
| c) $\lg n + 1$ | d) None |

Question 26

What will be the problem size of the recursive call for $T(n) = 2 T(n/2) + cn$ if $n = n/16$

- | | |
|------------|-------------|
| a) $n / 2$ | b) $n / 16$ |
| c) $n / 8$ | d) $n / 32$ |

Question 27

What will be the “problem size” at level “i” for $T(n) = 4T(2n/3) + O(n)$?

- | | |
|------------------|------------------|
| a) $n / 3^i$ | b) $n / 2^i$ |
| c) $n / (2/3)^i$ | d) $n / (3/2)^i$ |

Question 28

What will be the “cost of node” at level “i” for $T(n) = 4T(2n/3) + O(n)$?

- a) $cn / 3^i$ b) $cn / 2^i$
 c) $cn / (2/3)^i$ d) $cn / (3/2)^i$

Question 29

What will be the “number of nodes” at level “i” for $T(n) = 4T(2n/3) + O(n)$?

- a) 3^i b) 2^i
 c) 4^i d) $(3/2)^i$

Question 30

What will be the “cost of level” “i” for $T(n) = 4T(2n/3) + O(n)$?

- a) $cn (8 / 3)^i$ b) $4^i cn / 2^i$
 c) $3^i cn / (2/3)^i$ d) $2^i cn / (3/2)^i$

Question 31

What will be the “last level number” for $T(n) = 4T(2n/3) + O(n)$?

- a) $\log_4 n$ b) $\log_{2/3} n$
 c) $\log_{3/2} n$ d) $\log_3 n$

Question 32

What will be the “cost of last level” for $T(n) = 4T(2n/3) + O(n)$?

- a) $n \log_4 3/2$ b) $n \log_4 2/3$
 c) $n \log_{3/2} 4$ d) $n \log_{2/3} 4$

Question 33

What will be the value of $\sum_{i=0}^{\infty} x^i$ if $x < 1$

- a) $\frac{1}{1-x}$ b) $\frac{1}{1+x}$
 c) $\frac{1}{x-1}$ d) None

Question 34

In “Matrix Chain Multiplication”, for any matrix A_i , the order of the matrix is:

- a) $p_i * p_{i-1}$ b) $p_{i-1} * p_{i-1}$
 c) $p_{i-1} * p_i$ d) $p_i * p_i$ e) None

Question 35

The recurrence $T(n) \leq 2T(n/2) + \Theta(n)$ of Quick Sort algorithm reflects:

- a) Worst Case Partitioning b) Best Case Partitioning
 c) Balanced Partitioning d) None

Question 36

Which of the following condition is incorrect for “Master Theorem Method”?

- a) $\epsilon > 0$ b) $a f(n/b) \leq c f(n)$
 c) $c < 1$ d) None

Question 37

What is the running time of Quick Sort algorithm for “Balanced Partitioning”?

- a) n
- b) n^2
- c) $n \lg n$
- d) $\lg n$

Question 38

What is the running time of Quick Sort algorithm for “Best Case Partitioning”?

- a) n
- b) n^2
- c) $n \lg n$
- d) $\lg n$

Question 39

Suppose $A1 = 5 * 6$, $A2 = 6 * 8$, and $A3 = 8 * 9$, what will be the order of resultant matrix for optimal solution?

- a) $p_0 * p_1$
- b) $p_1 * p_2$
- c) $p_0 * p_3$
- d) $p_2 * p_3$
- e) $p_1 * p_3$
- f) None

Question 40

Suppose $A1 = 5 * 6$, $A2 = 6 * 8$, and $A3 = 8 * 9$, what will be the value of p_2 ?

- a) 5
- b) 6
- c) 8
- d) 9
- e) None

Question 41

The recurrence $T(n) = T(n-1) + \Theta(n)$ of Quick Sort algorithm reflects:

- a) Worst Case Partitioning
- b) Best Case Partitioning
- c) Balanced Partitioning
- d) None

Question 42

Suppose $A1 = 5 * 6$, $A2 = 10 * 5$, and $A3 = 6 * 5$, what will be the order of resultant matrix for optimal solution?

- a) $5 * 6$
- b) $10 * 5$
- c) $6 * 5$
- d) $10 * 6$
- e) None

Question 43

Suppose $A1 = 5 * 6$, $A2 = 10 * 5$, and $A3 = 6 * 5$, which 2 matrices will be multiplied first to achieve the optimal result?

- a) $A1$ & $A2$
- b) $A1$ & $A3$
- c) $A3$ & $A2$
- d) Can't be multiplied

Question 44

Suppose $A1 = 5 * 6$, $A2 = 10 * 5$, and $A3 = 6 * 5$, what will be the optimal scalar multiplications?

- a) 600
- b) 500
- c) 400
- d) 300
- e) None

Question 45

Which of the followings is common in both “Dynamic Programming” and “Divide and Conquer”?

- a) It stops the division when problem size is one
- b) Sub-problems solved separately
- c) Problem is divided into sub-problems
- d) b & c
- e) All of above

Question 46

Which of the followings is the characteristic specifically associated with “Divide and Conquer”?

- a) It stops the division when problem size is one b) Sub-problems solved repeatedly
- c) Problem is divided into sub-problems d) b & C e) All of above

Question 47

Which of the followings is the technique which may be used to solve a problem having multiple solutions?

- a) Divide and Conquer b) Dynamic Programming
- c) Brute Force Algorithm d) b & C e) All of above

Question 48

How many steps are involved in computing the “Optimal Value” using “Dynamic Programming”?

- a) 2 b) 3
- c) 4 d) None

Question 49

Which of the following steps is optional in “Dynamic Programming”?

- a) Characterize the structure of Optimal Solution
- b) Recursively define the value of an Optimal Solution
- c) Construct the Optimal Solution from computed information
- d) None

Question 50

Which of the followings is also known as “Optimization” technique?

- a) Divide and Conquer b) Dynamic Programming
- c) Brute Force Algorithm d) b & C e) All of above

Question 51

With respect to “Assembly Line Scheduling”, which option is correct?

- a) Skipping the station b) Vertical Transfer
- c) Re-ordering the stations d) Find minimum cost e) All of above

Question 52

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for $S_{i,j}$?

- a) Cost of station j of line i b) Cost till station j of line i
- c) Station j of line i d) Station i of line j e) None of above

Question 53

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for $a_{i,j}$?

- a) Cost of station j of line i b) Cost till station j of line i
- c) Cost of station i of line j d) Cost till station i of line j e) None of above

Question 54

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for $t_{i,j}$?

- a) Transfer cost from station j to line i b) Transfer cost to station j from line i
- c) Transfer cost to station j to line i d) Transfer cost from station j from line i e) None

Question 55

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for e_i ?

- a) Entry cost from line i b) Exit cost to line i
c) Exit cost from line i d) Entry cost to line i e) None

Question 56

With respect to “Assembly Line Scheduling”, which option describe the purpose correctly for x_i ?

- a) Entry cost from line i b) Exit cost to line i
c) Exit cost from line i d) Entry cost to line i e) None

Question 57

Which value is calculated by $f_2[j-1] + t_{2,j-1} + a_{1,j}$?

- a) $S_{1,j}$ b) $S_{2,j}$
c) $S_{2,j-1}$ d) $S_{1,j-1}$ e) None

Question 58

Suppose $f_1[n] = 16$, $f_2[n] = 20$, $x_1 = 5$ and $x_2 = -7$ what will be the value of “ l^* ”?

- a) 1 b) 2
c) both a and b d) None

Question 59

Suppose $f_1[n] = 16$, $f_2[n] = 20$, $x_1 = 5$ and $x_2 = -7$ what will be the value of “ f^* ”?

- a) 21 b) 27
c) 13 d) 25 e) None

Question 60

Suppose $A1 = 5 * 6$, $A2 = 10 * 5$, and $A3 = 6 * 5$, which sequence will make a “Chain”?

- a) A1 A2 A3 b) A2 A1 A3 c) A3 A1 A2
d) A1 A3 A2 e) A2 A3 A1 f) A3 A2 A1
g) None

Question No. 1: (Linear Time Sorting)**10 points**

Determine the expected running for the “Bucket Sort” algorithm. The algorithm is given for your reference.

BUCKET-SORT(A)

```
1  let  $B[0 \dots n - 1]$  be a new array
2   $n = A.length$ 
3  for  $i = 0$  to  $n - 1$ 
4      make  $B[i]$  an empty list
5  for  $i = 1$  to  $n$ 
6      insert  $A[i]$  into list  $B[\lfloor nA[i] \rfloor]$ 
7  for  $i = 0$  to  $n - 1$ 
8      sort list  $B[i]$  with insertion sort
9  concatenate the lists  $B[0], B[1], \dots, B[n - 1]$  together in order
```

Solution:

Question No. 2: (Recursion Tree Method)**15+5 points**

Solve the following recurrence using Recursion Tree method and prove it using “Substitution Method”.

$$T(n) = 3T(\lfloor n/4 \rfloor) + \Theta(n^2)$$

Solution:

Question No. 2: (Recursion Tree Method)

Question No. 2: (Recursion Tree Method)

Question No. 3: (Master Theorem Method)

10 points

Solve the following recurrence using Master Theorem method.

$$T(n) = 3T(n/4) + n \lg n$$

Solution:

Extra Sheet for Rough Work